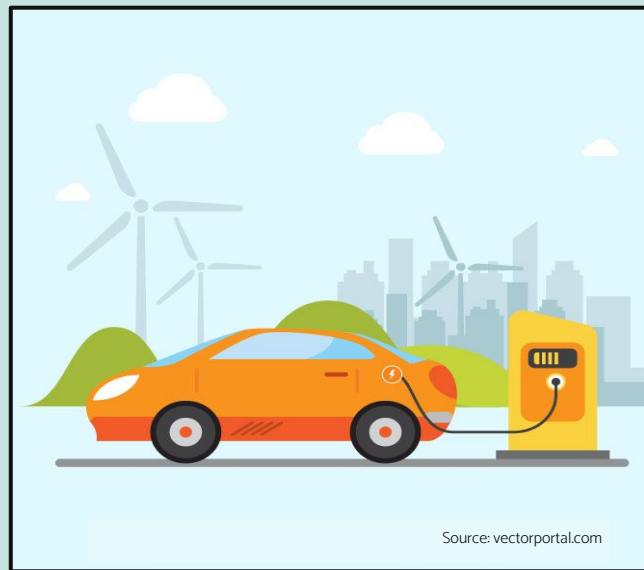
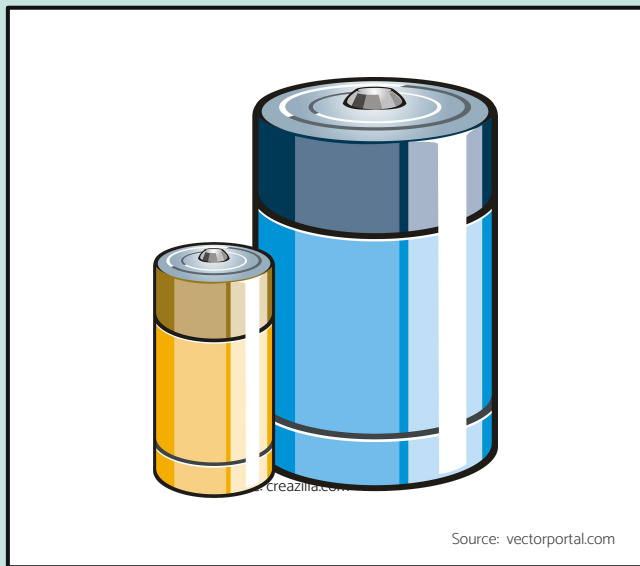


THE LITHIUM CONVERGENCE

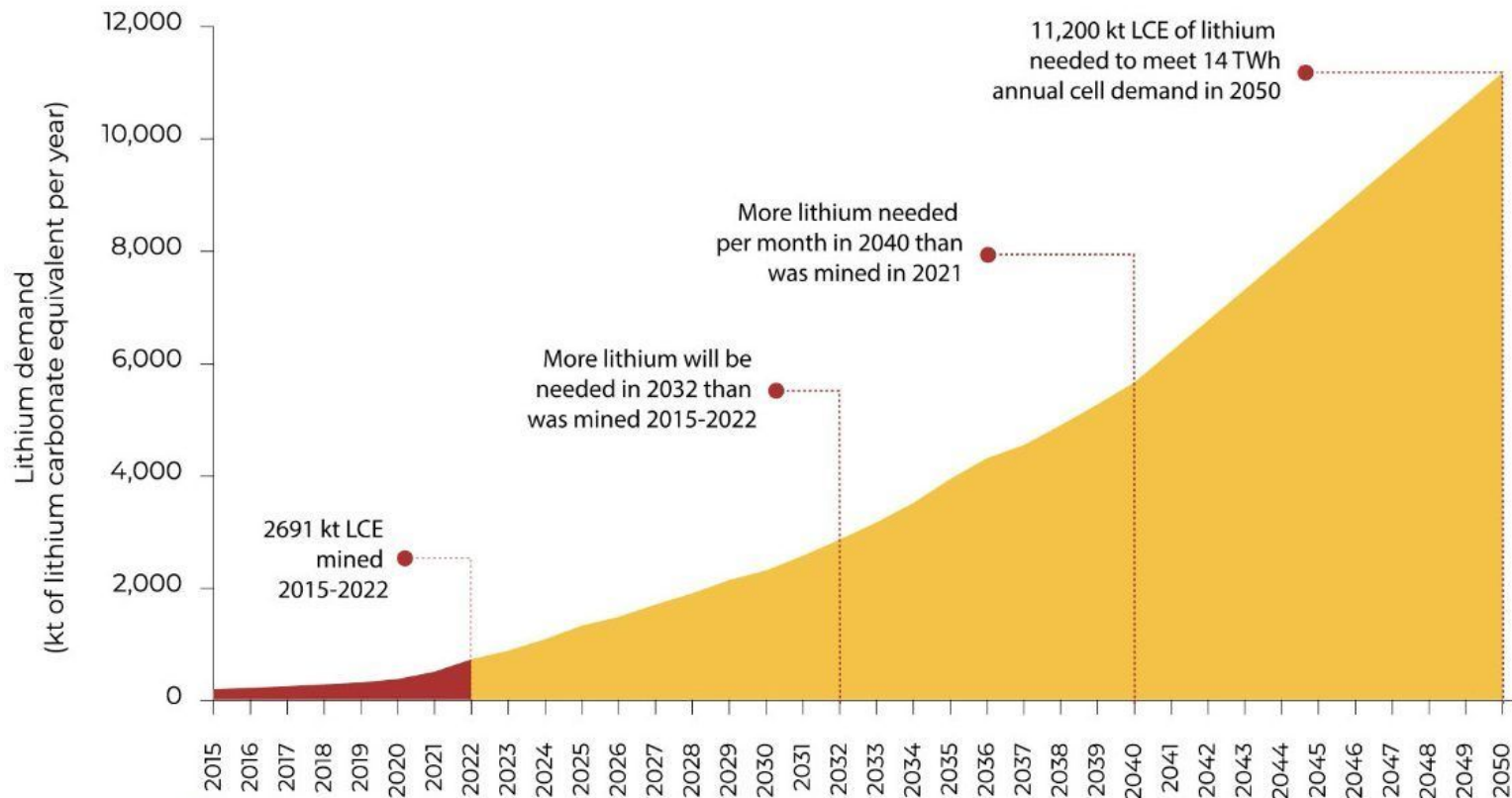
Sustainably Balancing Lithium Supply and Demand.

Source: NASA/Wikimedia Commons

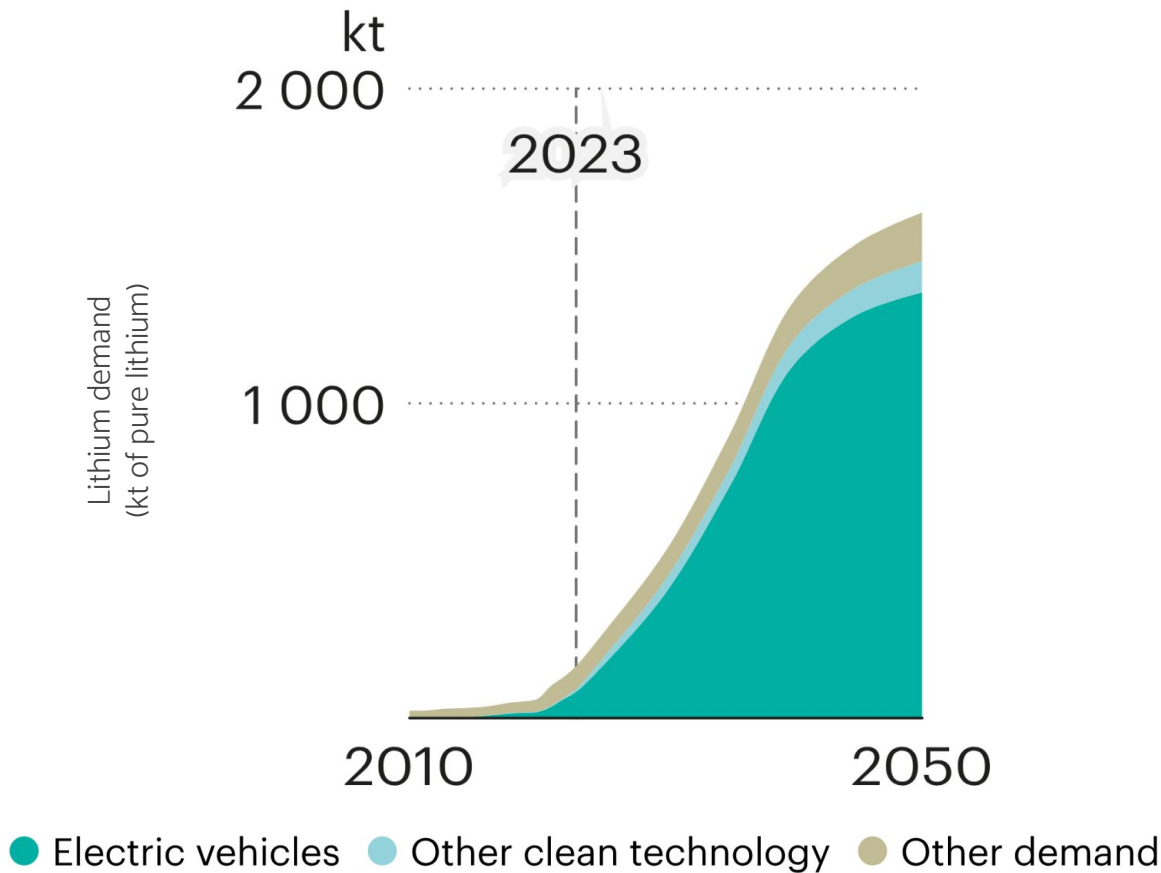
Let's Talk Lithium.



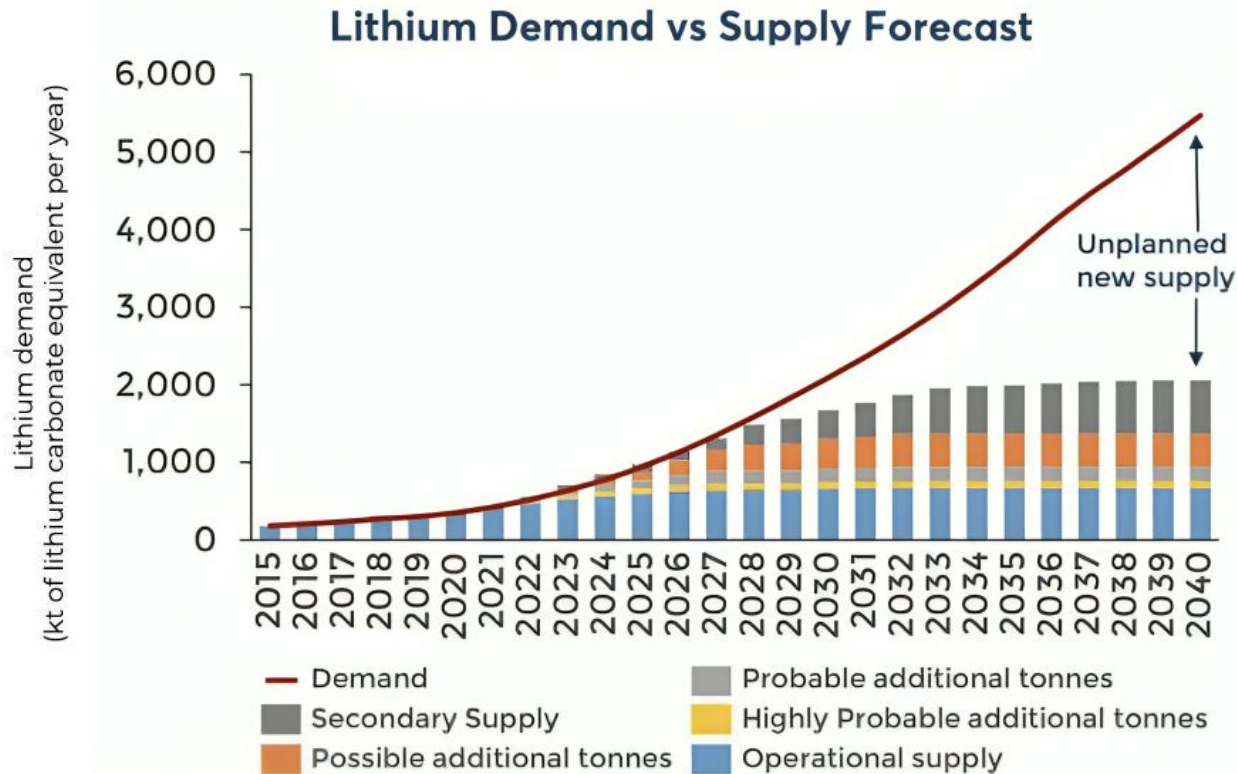
Lithium demand will rapidly increase due to renewable energy storage demands



The majority of lithium demand comes from electric vehicles



However, lithium supply is not projected to reach those demands



Source: Benchmark Minerals

Mining lithium also has significant social and environmental impacts



A lithium mine in Chile.

Source: People's Dispatch

Problem Statement Summary

The transition to a carbon-neutral economy heavily depends on the availability of lithium. There are lithium deposits in the U.S., but mining has substantial local impacts, especially on Indigenous communities.

How can the gap between expected supply and demand of lithium in the U.S. be bridged, while minimizing the environmental and social harm that current production methods cause?

THE CURRENT STATE

How is lithium being used now?

REDUCING DEMAND

How can we shift away from using lithium?

REFORMING THE LITHIUM SUPPLY CHAIN

How can we sustainably produce more lithium?

IMPLEMENTATION

How do we converge the supply and demand of lithium?

Our Solution: A Multi-Step Policy Proposal

Following **social impact guidelines...**

Policies to Implement by 2030

- **Prioritize** Direct Lithium Extraction and developing sustainable and efficient mining technology
- **Incentivize** iron-air and sodium batteries
- **Begin** investing in battery-electric buses

Policies to Implement by 2050

- **Install** 40-70 lithium recycling plants
- **Build** out public transit infrastructure and pilot project microtransit
- **Invest** in upcoming energy storage methods

How We've Approached Social Impact

- 1) Keeping social impact, particularly on local and indigenous communities, in mind **every step of the way.**
- 2) We looked to find community engagement solutions **largely supported by** indigenous community groups.
- 3) The policy proposals we present today are the ones we have found to avoid the most social and environmental harm .
 - a) Considering climate change disproportionately harms the vulnerable, this includes ensuring an equitable renewable energy transition.
- 4) Every project we propose will need to follow our established social impact criteria.

Social Impact Criteria

- 1) Free, Prior, and Informed Consent of Indigenous communities
- 2) Ongoing conversations with community members/stakeholders
 - a) Goal of developing a long-term relationship with community
 - b) What benefits can the project provide?
 - c) What concerns should be addressed?
- 3) Robust Environmental Impact Assessments
- 4) Priority to community-led projects



The Current State of Lithium

01

Source: Steve Morfi/ Shutterstock.com

Current Mining Technology: Hard Rock & Brine Mining

TRADITIONAL MINING



Hard rock mining

Greenbushes Lithium Mine, AU. Source: Talison Lithium

Brine mining



Silver Peak Lithium Mine, Nevada, US. Source: Wikipedia

Hard Rock Mining: Extraction and Refinery

Extraction



Types of hard rock: spodumene & lepidolite
Greenbushes Hard Rock Mine, AU
Image source: Talison Lithium

Refining

1. Crushing
2. Separation
3. Floatation
4. Roasting
5. Sulfuric Acid
6. Cooled
7. 2nd Roasting

Purification



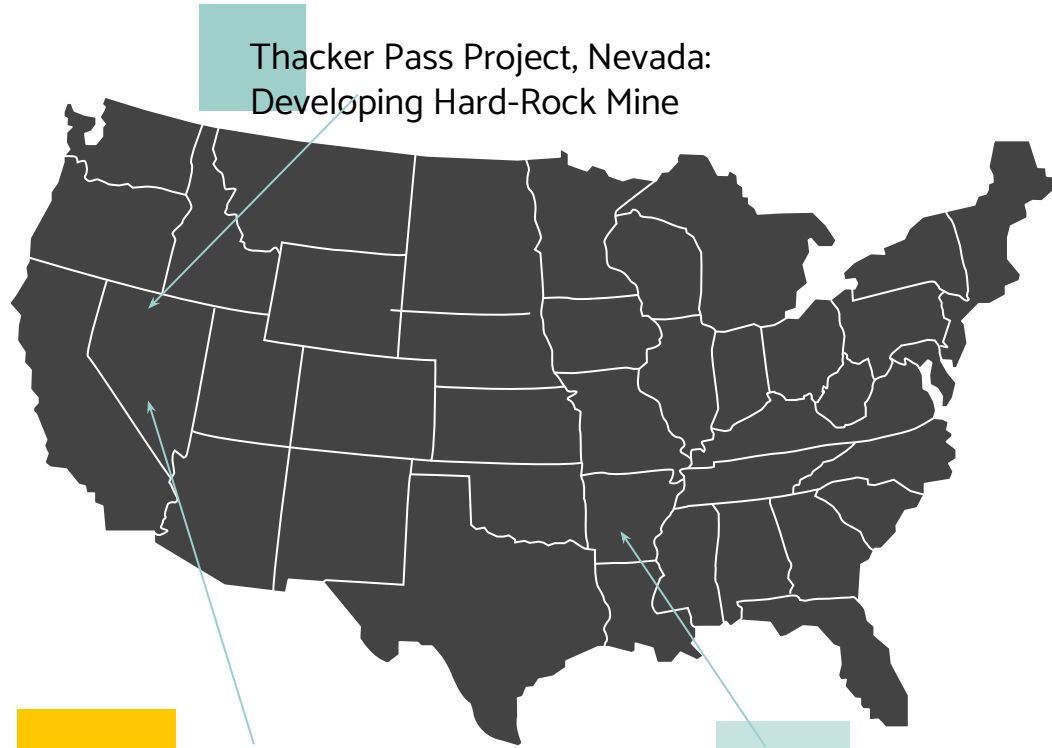
Lithium Sulfate
Image source: RecycLiCo

Hard Rock Mining: Environmental & Social Impact



Source: BC Gold Adventures

- Carbon Emissions
15 million kg of CO₂ footprint
per kt of Lithium
- Land Footprint
115 acres needed per kt Lithium
- Tailings
Sulfuric Acid
- Dust & Air Quality



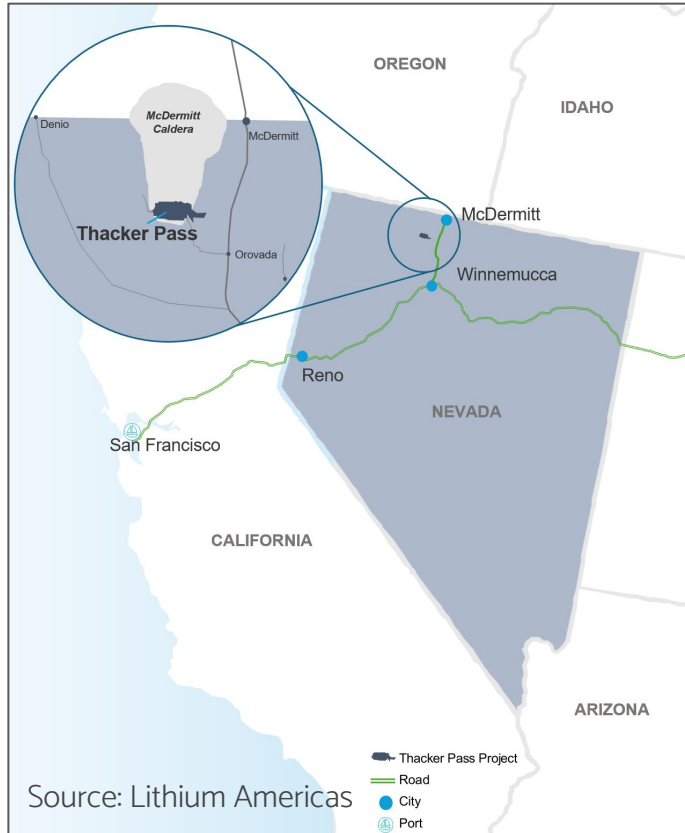
Thacker Pass Project, Nevada:
Developing Hard-Rock Mine

Albemarle Silver Peak Mine, Nevada:
Operating Brine Mine

Smackover Formation Project, Arkansas:
Developing Brine Mines

Current Lithium Mines + Projects in the US

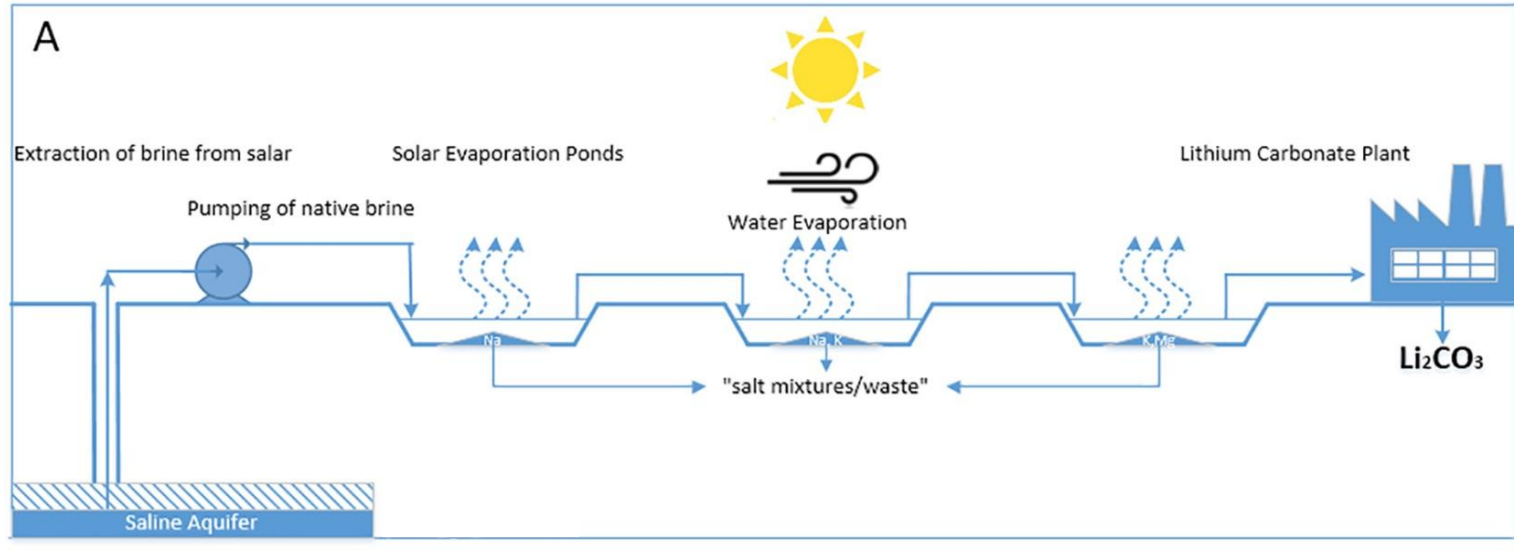
Social Impacts of Hard Rock Mining: Thacker Pass



- Located on the massacre site of the Numu people and sacred to many surrounding Indigenous groups
- Multiple lawsuits filed opposing the mine based on threats to water, wildlife and cultural aspects that have lasted up to years

Solution: Move away from hard rock mining to avoid infringing on communities and for faster mining implementation

Brine Mining: Extraction and refinery



Silver Peak, Nevada Brine Mine:

550,000 Gallons of Water / Ton of Lithium

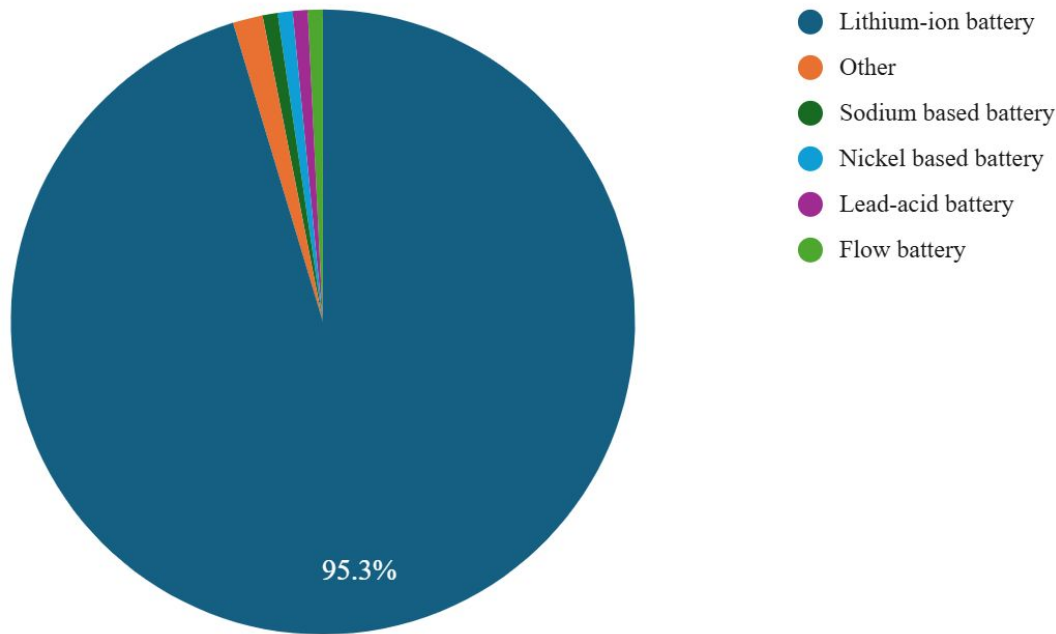
Source: Flexer et. al

Current Energy Storage

Grid-Scale

Lithium-ion batteries form majority of U.S. utility-scale storage

Primary Utility-Scale Energy Storage Methods in U.S.



Composition of energy storage methods on the utility scale in the U.S. - Source: U.S. Energy Information Administration, Form EIA-860, 2023 Annual Electric Generator Report

Why use Lithium for grid storage?

Low Cost

\$139/kwh

High Energy Density

**200-300
Wh/kg**

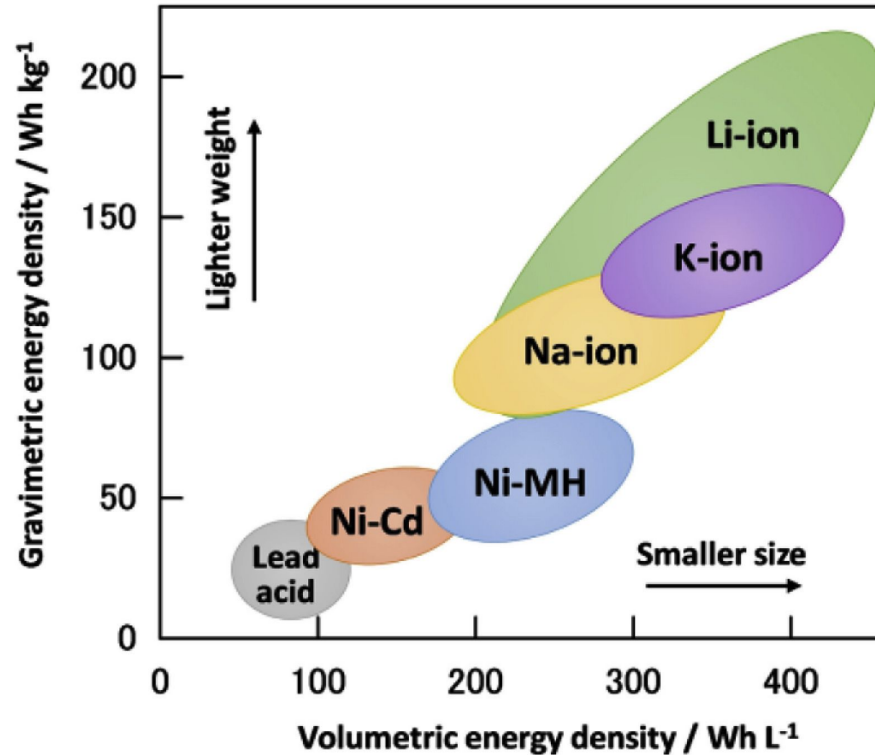
Long Cycle Life

**1000 -
5000
cycles**

Current Energy Storage

Mobile-Scale

Lithium-ion batteries are lightweight and small compared to the energy they store.



Energy densities of common mobile-scale batteries. Source: El Kharbachi

Why use Lithium for mobile storage?

Lightweight

200-300
Wh/kg

Compact

300-400
Wh/L

Long Cycle Life

1000 -
5000
cycles

Current Lithium Demand

Lithium Demand for EV's

Why go electric?

Transportation accounts for 28% of U.S. greenhouse gas (GHG) emissions; 80% from on-road vehicles

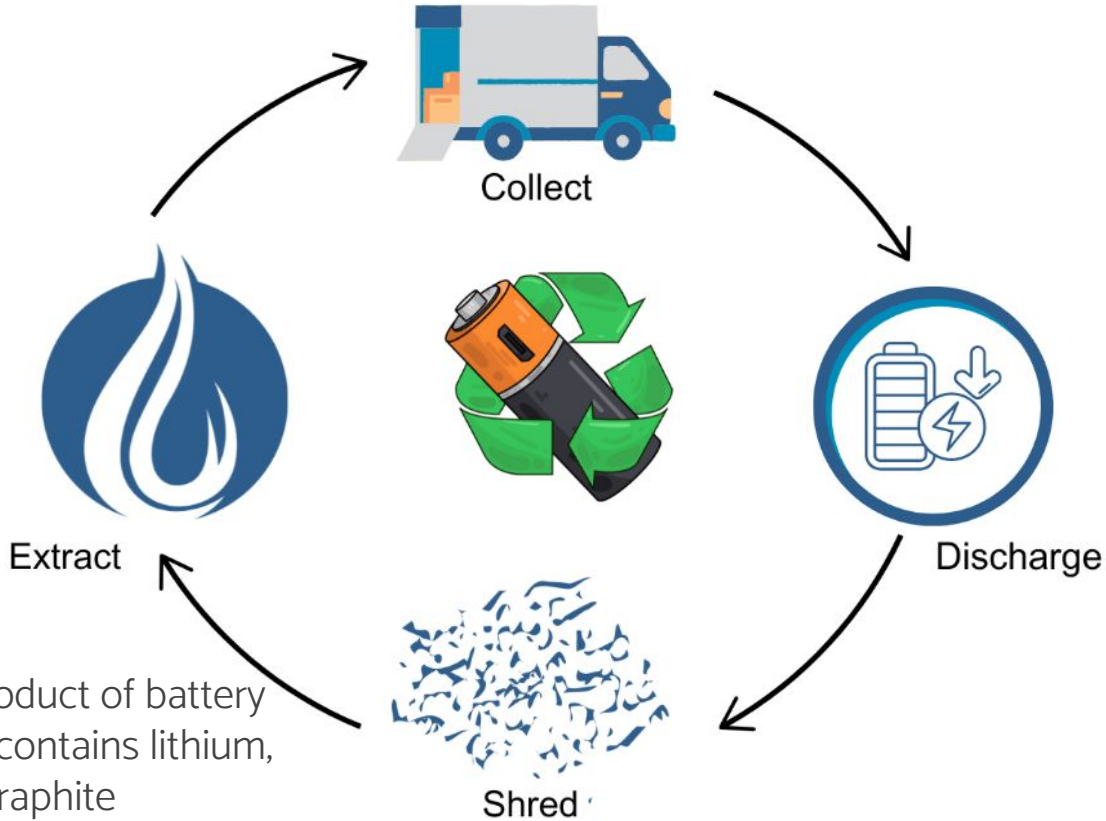
The Challenge

Electrification of personal vehicles requires ~1.9 billion kg of lithium by 2050

Going electric doesn't work

Lithium demand is projected to exceed supply by 300 million kg by 2030

Battery Recycling



Black mass: product of battery shredding that contains lithium, cobalt, nickel, graphite

Image source: Arvia Technology

Current Recycling Methods

Pyrometallurgy

- Smelting process: shredded battery parts are melted at $\sim 1000^{\circ}\text{C}$
- Recovers lithium and cobalt
- Produces toxic gases (dioxins and furans) from combustion of plastic components
- 4,212 kg CO_2 per ton of black mass



Pouring metal into a metallurgical electric arc furnace.

Source: Aqua Metals

Current Recycling Methods

Hydrometallurgy

- Black mass is dissolved in acid, then metal ions are removed chemically
- Recovers lithium, cobalt, nickel, and manganese
- Produces contaminated wastewater
- 2,960 kg CO₂ per ton of black mass





Pouring metal into a metallurgical electric arc furnace.

Source: Aqua Metals

Pyrometallurgy

- Recovers lithium and cobalt
- Produces toxic chemicals (dioxins and furans) from incineration of plastic components
- 4,212 kg CO₂ per ton of black mass

Hydrometallurgy

- Recovers lithium, cobalt, nickel, and manganese
- Produces contaminated wastewater
- 2,960 kg CO₂ per ton of black mass



Mercedes Benz's recycling facility in Kuppenheim, Germany.

Source: Mercedes Benz

Go electric - go carbon neutral

The problem

Transportation accounts for 28% of U.S. greenhouse gas (GHG) emissions; 80% from on-road vehicles

The solution

The Biden-Harris administration has set a goal to “go green” - electrify personal vehicles

We can't go electric

Demand: 500 million kg
annually by 2030



Electrification of PV ~240million
vehicles *8kg = 1.9billion kg of
lithium by 2050

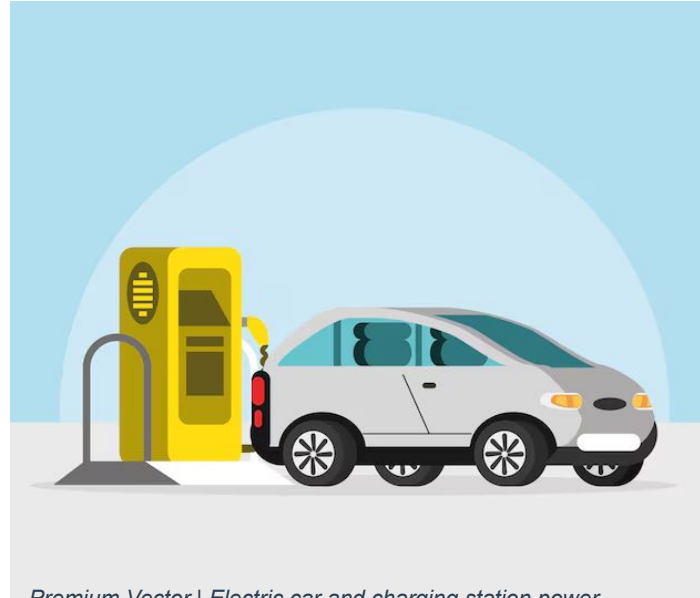
Access to clean transportation isn't equal for everyone

- Electrical vehicles are still too expensive for lower-income families



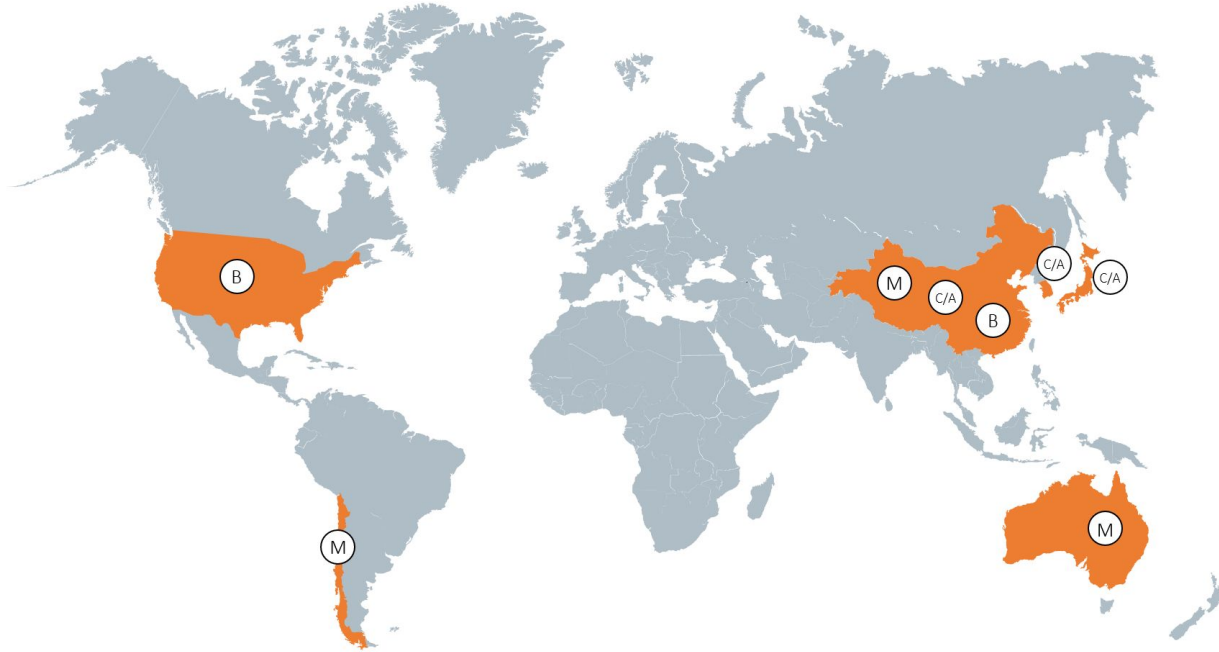
YouTube not suitable.png - Wikimedia Commons.

- Lack of charging stations



Premium Vector | Electric car and charging station power.

From mines to batteries



70% of lithium mined in Australia, China, Chile

80% of batteries manufactured in China

Lithium considered to have **high price volatility** by IEA

A world map highlighting major countries in each stage of the supply chain.

M - mining, **C/A** - refining lithium into **cathode/anodes**, **B** - battery production



Reforming the Lithium Supply Chain

02

Source: Steve Morfi/ Shutterstock.com



DOMESTIC SUPPLY CHAIN

Improving
stability of
lithium supply

MINING

The future of
lithium mining:
Direct Lithium
Extraction (DLE)

RECYCLING

New technologies
causing reduced
environmental and
social harm

The Case for a Domestic Lithium Supply Chain

A Domestic Lithium Supply Chain

Why?

1. Lessens the effects of global market disruptions domestically.
2. Potentially improves global lithium markets.

However,

1. It is critical to make sure these developments benefit and not harm communities on a local scale.
2. Need to carefully consider effect on international relations.

The Future of Lithium Mining: Direct Lithium Extraction (DLE)

Direct Lithium Extraction

Mining lithium directly from the brine water through chemical and mechanical methods

TRADITIONAL



Silver Peak Lithium Brine Mine, Nevada, US.
Source: Wikipedia

DLE



DLE Facility
Source: Sunresin

DLE vs Traditional Mining Methods

LCE = Lithium Carbonate Equivalent

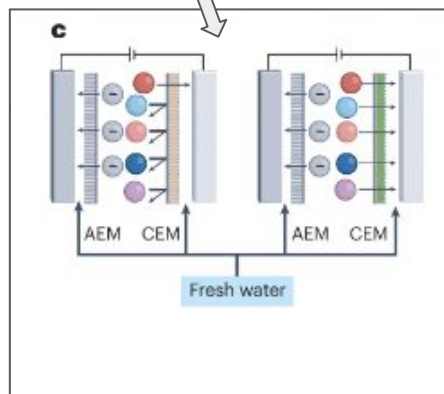
A uniform measurement of the lithium since it comes in different forms

Source: Lithium Harvest

	DLE from Brine	Solar Evaporation Brine Extraction	Hard Rock Mining
Project Implementation Time	5-7 years	13-15 years	8-10 years
Lithium Carbonate Production Time	2 hours	2-3 years	3-6 months
Lithium Yield	80-95%	20-40%	6-7%
Average Footprint per 1000 mt of LCE	1.4 acres	65 acres	115 acres
Water Usage per 1000 mt of LCE	80 million gallons	550 million gallons	250 million gallons
CO2 Footprint per 1000 mt of LCE	1.5 million kg	5 million kg	15 million kg
Avg. Cost per mt of LCE	\$5,700	\$5,800	\$6,900

Direct Lithium Extraction (DLE) Methods

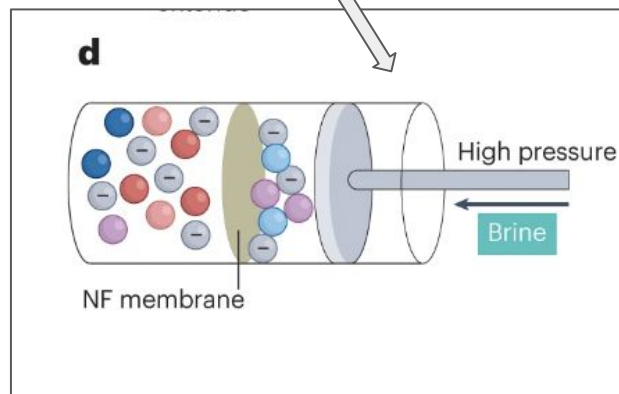
Electromembrane Selection



AEM/CEM = Anion/Cation
Exchange Membrane

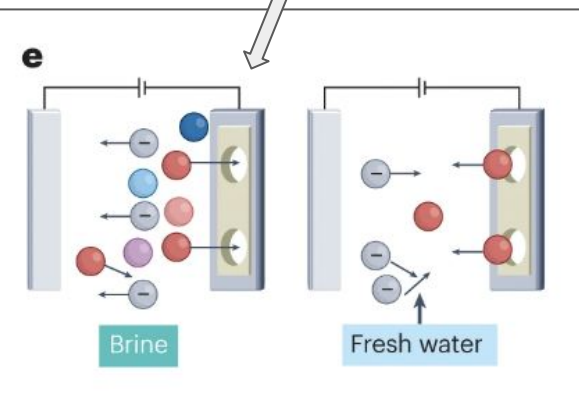
Source: Vera, et al

Mechanical Filtration

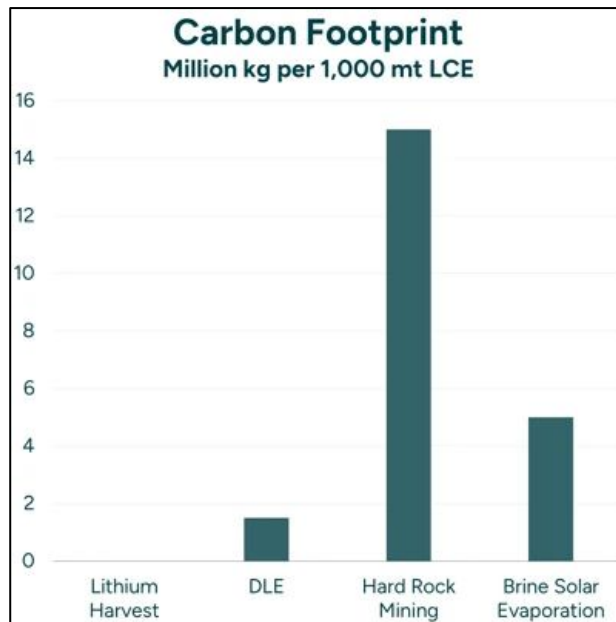
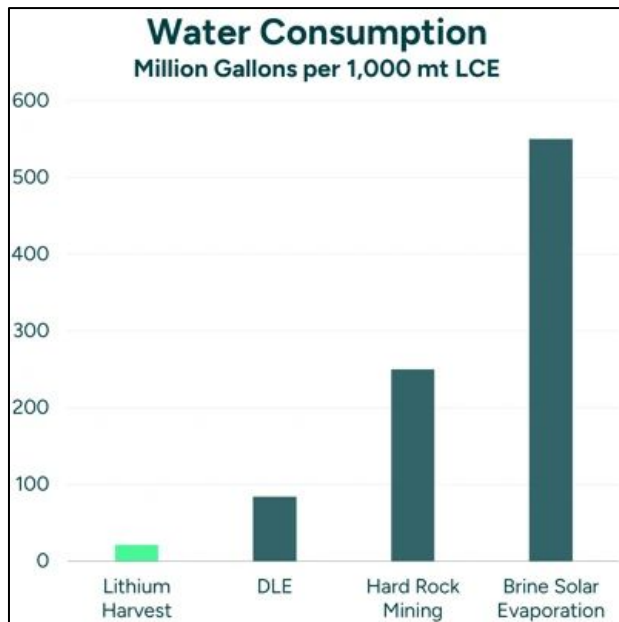


NF Membrane: Nanofiltration Membrane

Electrochemical Filtration



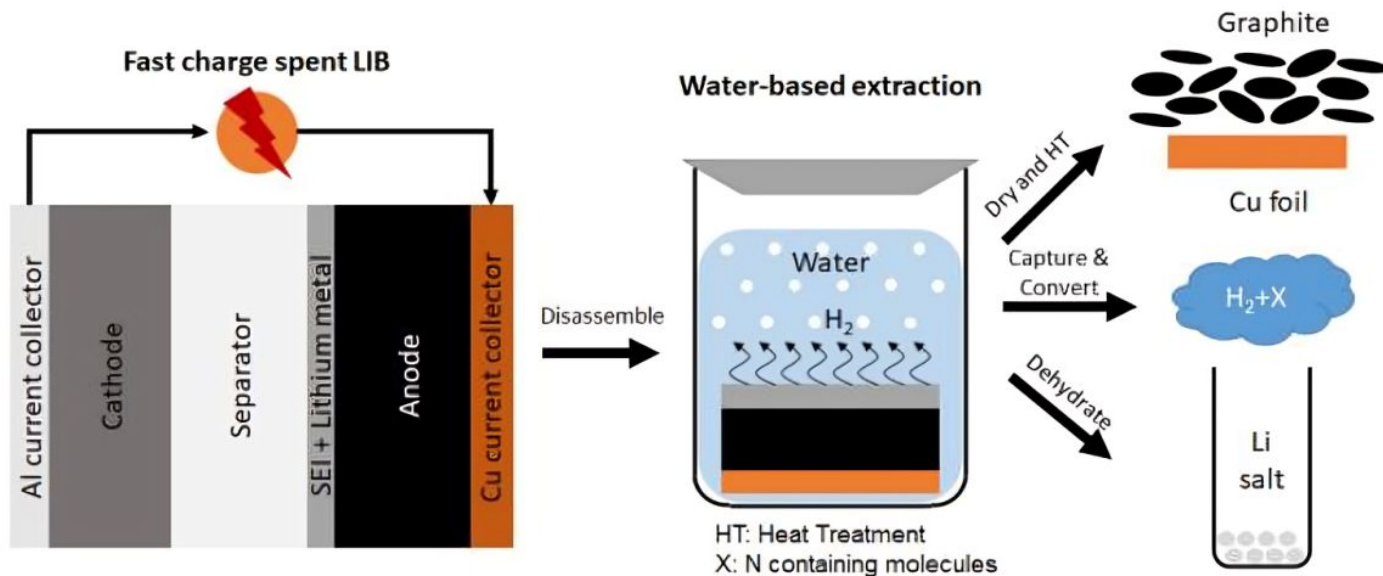
Lithium Harvest Method: Oil-produced water



- requires **12-15 months to set up** next to existing mainstream companies and SWDs vs. **5-7 year** implementation time of other DLE

Source: Lithium Harvest

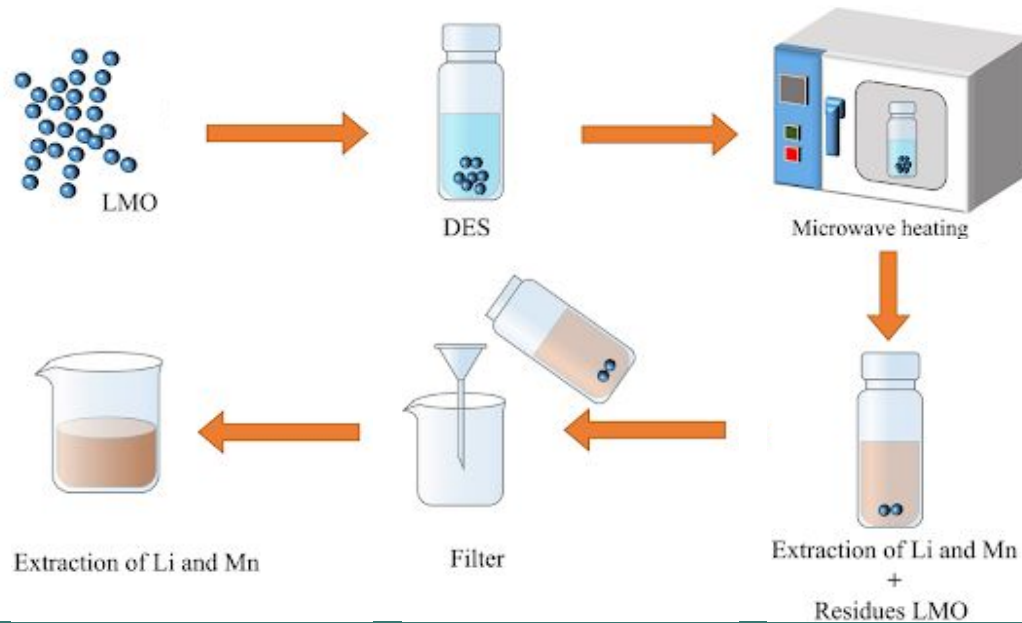
New Recycling Technologies: BRAWS and MADES



BRAWS

Battery Recycling And Water Splitting

- Battery disassembly is currently a challenge.
- Up to 92% of recovery with 97% purity at a lab scale.
- The hydrogen produced can be captured and sold as fuel.



MADES

Microwave-Assisted Deep Eutectic Solvent
Technology

- DES is biodegradable and safe to handle.
- Nearly 100% recovery in 30 minutes and 97% purity at a lab scale
- DES can be modified to target other metals.



Reducing Lithium Demand

03

Source: Steve Morfi/ Shutterstock.com

Alternative Energy Storage

Short-Term (2030)

Iron-air, Geothermal, Sodium

Short Term - Grid-Scale

Iron-air

- Iron is very abundant
- Iron is commonly recycled

1,080
Wh/kg

\$20
per
kWh

Lithium-ion

- Lithium is a “Critical mineral”
- Lithium is less commonly recycled

200-
300
Wh/kg

\$139
per
kWh

Short Term - Grid-Scale

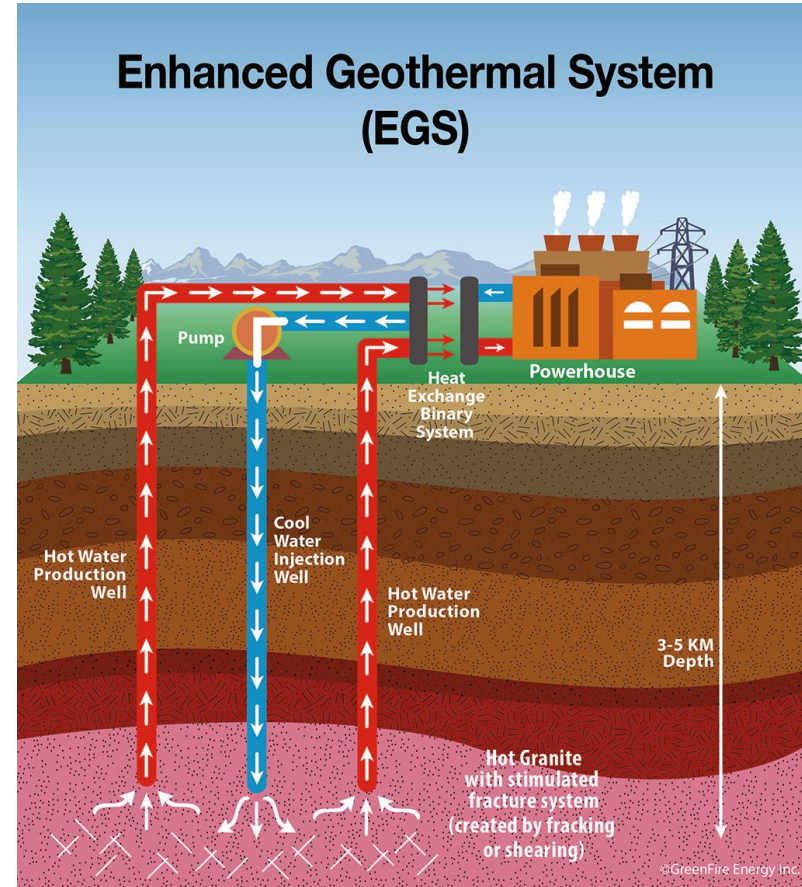
Geothermal Energy Storage

Geopressedured Geothermal System

- High upfront cost
- \$0.07/kWh after construction is finished
- Repurposes fossil fuel facilities

Enhanced Geothermal System

- High upfront cost
- \$0.045/kWh after construction is finished



Short Term - Mobile-Scale

- Intended as **additional option** for electric vehicle batteries

	Energy density (Wh/kg)	Lifetime (cycles)
Lithium	100–300	1000–5000
Sodium	100–160	4000

\$10
per
kWh

Comparison of Lithium and Sodium-ion batteries. Source: American Physical Society.

Note: relevant data points taken from larger chart for clarity on-screen

- Sodium collected through electrolysis

Alternative Energy Storage

Long-Term (2050)
Hydrogen Fuel Cell, Solid-State

Long Term Grid-Scale

- Hydrogen Fuel Cells
 - Extremely energy dense -
33 kWh/kg
 - Environmentally friendly
 - Catalyst is expensive

Grey vs. Green Hydrogen

**\$1.50
per
kg**

**\$7.00
per
kg**

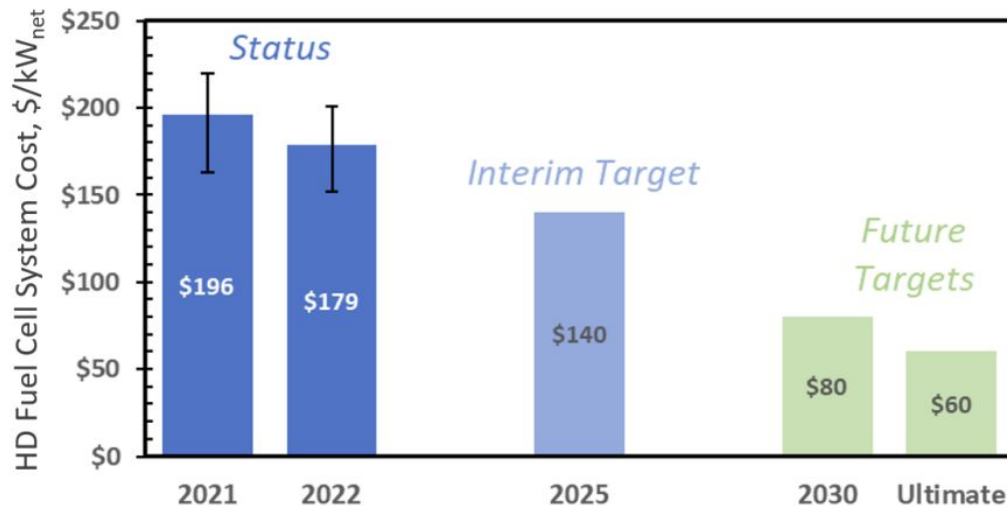


Figure 2: Modeled HD Fuel Cell System Cost Status (2021, 2022) and Interim Target (2025) for a manufacturing volume of 50,000 systems per year. Future (2030, \$80/kW), and ultimate (\$60/kW) targets at 100,000 systems per year.¹²

Source: Department of Energy

- Further development of geothermal

Long Term Mobile-Scale

Current Solid-State

- Early phase deployment
- Lithium in anode and/or cathode
- High energy density

~500 Wh/kg

Lithium-free Solid-State

- Research-stage development
 - Potassium and sodium silicate
 - Transition Metal Oxide
- High theoretical energy density

~1000 Wh/kg

Public Transit:
Reducing personal EV Demand
while meeting renewable energy goals

Reducing Lithium Demand

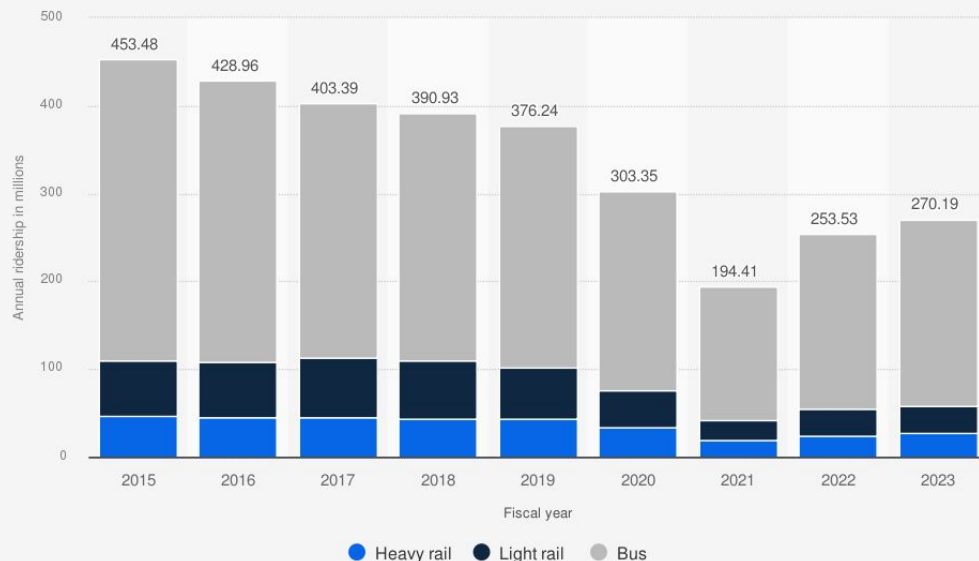
Reduce the number of
personal electric vehicles
on the road

Use sodium-ion
batteries in EVs instead
of lithium

Benefits of Bus Public Transportation

Buses are the main form
of public transit in some
American cities

Annual number of passenger boarding of Los Angeles County's transit authority (LACMTA) from FY 2015 to FY 2023, by mode (in millions)

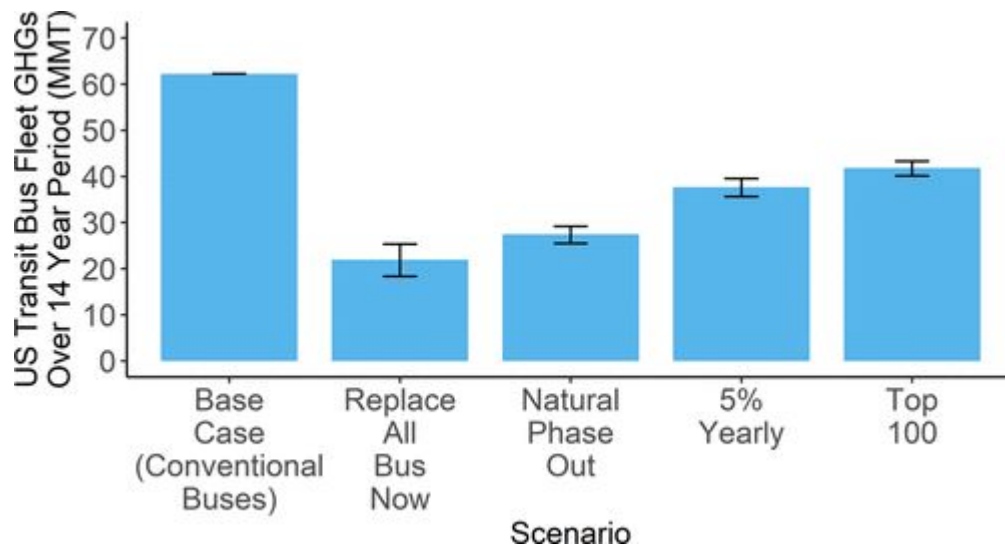


Source: Metro in Statista

Electrifying public transport reduces emissions

This is where
electrification comes in!

Start with bus
electrification



Source: Martinez and Samaras

How much lithium is needed for bus electrification?

For EVs: 1.9 billion tons of lithium
to electrify 240 million vehicles

~68,500 buses
in US fleet

X

~40-100 kg
lithium per bus

=

2.7-7.2 million
kg lithium for
US buses

Using Sodium-ion Batteries in EVs

Transition EVs to use
sodium-ion batteries
instead of lithium



CATL

Addressing Rural Communities

Microtransit:

Providing
on-demand
transport with
flexible routes

Optimal for the
varying travel
patterns of rural
communities

Vehicles are smaller
and easier to electrify

Implementation

04

Source: Steve Morfi/ Shutterstock.com

What Does Successful Community Engagement Look Like?



Engagement Principles	
Meaningful and Ongoing Relationships	
Cultural Connection	
Long-Term Financial Opportunities	
Capacity Development	

Mining Policy

Focusing on Direct Lithium
Extraction

DLE: Environmental and Social Impact Policy

1. Freshwater Usage

- Electrical and mechanical filtration: can use ten times less water, they use approximately one hundred times more energy than existing mining methods
- Focus on lithium deposits in less water scarce regions i.e Smackover Formation Project

2. Waste Brine Disposal

- Utilizing existing monitoring systems for oil drilling in DLE water return
- Less chemically intense extraction methods: **reverse osmosis and nanofiltration**

3. Education & Agency

- More peer-reviewed and digestible articles surrounding DLE technology and environmental and social implications

DLE Implementation Goals

Short Term (2030):

1. Legal agreements for discontinued use of DLE from oil produced water as the U.S moves away from oil
2. Enhanced water management and regulatory measures in existing mines i.e Silver Peak Brine Mine, Nevada

Long Term (2050):

1. Mechanical & Electrochemical filtration
2. Tax subsidy to incentivise DLE development

Energy Storage Policy

Energy Storage Policy

\$662 billion USD

U.S. Implicit & Explicit
Fossil Fuel Subsidies

\$3 billion USD

2024 US Battery
Production Investment

2.3 tons

Amount of carbon produced
for every ton of lithium at
Thacker Pass

Energy Storage Policy

Short Term (2030)

- Redirect \$3 billion from portions of lithium and fossil fuel subsidies
- Invest into iron-air batteries, sodium batteries, and geothermal storage
 - Grants and subsidies

Long Term (2050)

- Continue redirection of funds
- Invest further \$3 billion into hydrogen fuel cell and solid-state batteries

Recycling Policies

RECYCLING POLICY- Short Term (2030)

- 1.** Subsidizing recycling plant repurposing for MADES and BRAWS.
- 2.** Implementing a \$500 tax credit per ton of batteries with 7% recycled lithium to manufacturers.

RECYCLING POLICY- Long Term (2050)

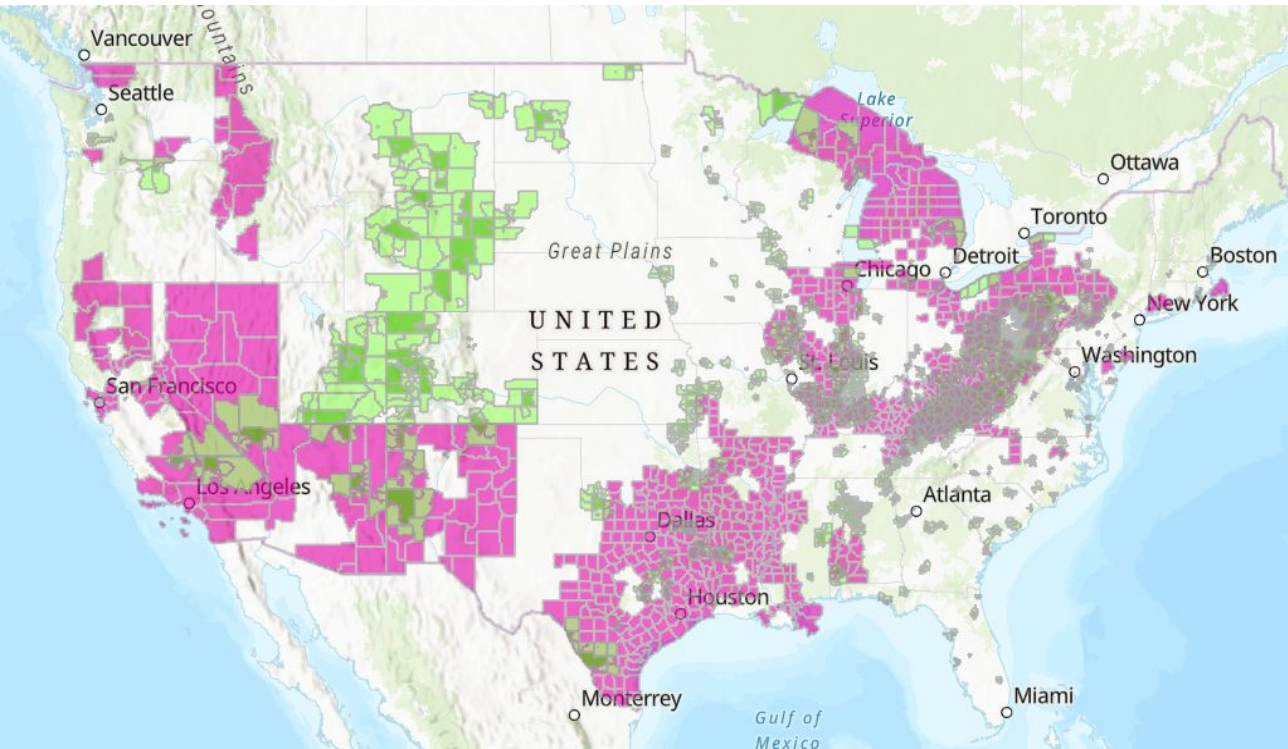
- Transitioning to a circular economy
- Expansion of recycling infrastructure
- 40-70 new facilities

**Reduce
transportation
costs**

**Identify funding
availability**

**Minimize local
community and
environmental
impact**

RECYCLING POLICY- Long Term (2050)



-  2024 Coal Closure Energy Communities
-  2024 Other Energy Communities

Energy communities in the contiguous 48 states (U.S. Department of Energy, 2024)

Public Transportation Policies

Public Transportation - Short Term (2030)

Promote
alternatives to
lithium-ion
batteries for EVs

Electrify buses
through a 14-year
phase out

Electrifying Buses

22% of US buses are eligible for replacement

\$8.5 billion to replace these buses with battery-electric buses

Replace diesel buses as they reach end-of-life annually for \$2-3 billion per year

Public Transportation - Long Term (2050)

Provide grants
for regions to
expand public
transit

Begin pilot
projects for
microtransit

The background of the slide is a photograph of a rocky, uneven terrain. In the foreground, there are numerous grey and brown rocks of various sizes. In the background, a body of water is visible, and the sky is a pale blue. A semi-transparent green rectangular box is positioned in the upper left quadrant of the image.

Conclusion

05

Short Term (by 2030):

- 1) Switching to more efficient and sustainable mining methods by prioritizing subsidies of Direct Lithium Extraction (DLE)
- 2) Expanding immediately viable alternative energy storage solutions by incentivizing iron-air and sodium batteries
- 3) Subsidise MADES and BRAWS recycling methods and implement tax credit for batteries using 7% recycled lithium
- 4) Beginning to invest in battery-electric buses in order to eventually lighten lithium demand for personal EVs

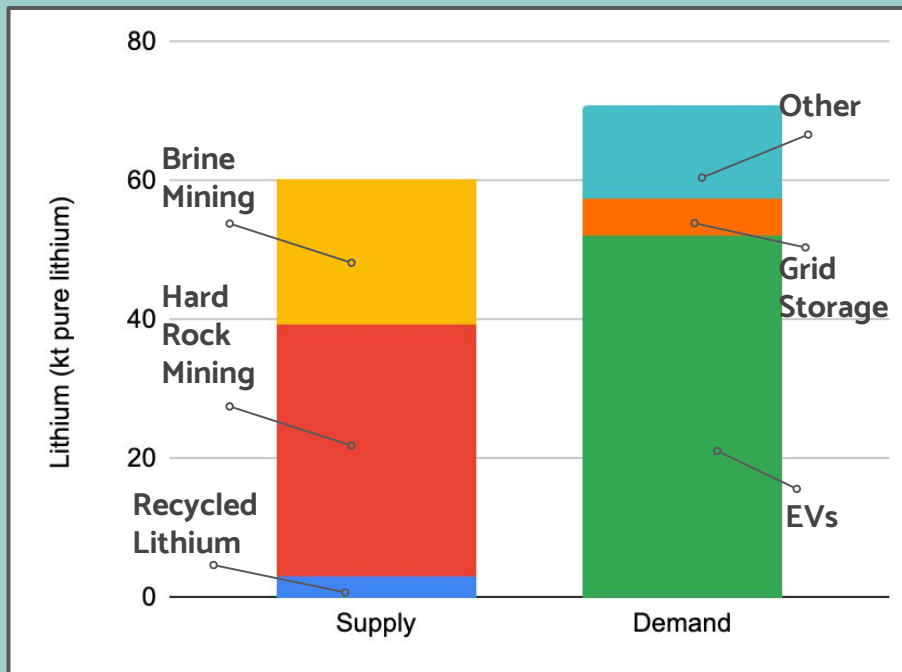
Long Term (by 2050):

- 1) Installing 40-70 recycling plants to move to a circular lithium economy and decrease dependence on lithium mining
- 2) Building out additional public transit infrastructure and initiating pilot projects for microtransit to reduce lithium demand for personal EVs
- 3) Investing in alternative energy solutions that show potential to be effective in the long term - hydrogen fuel cell and solid-state

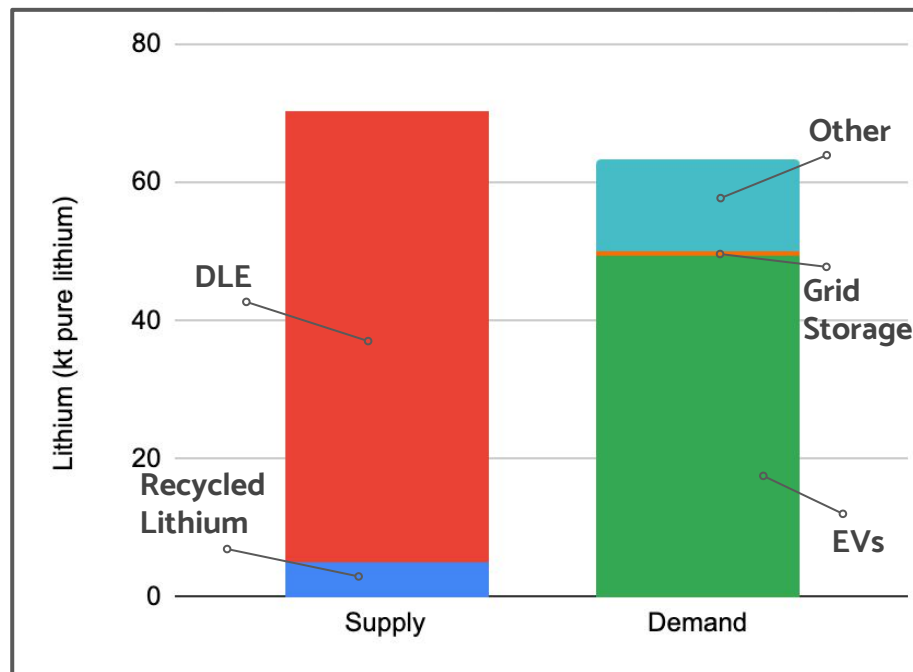
During both the short and long term, projects must follow social impact criteria.

Current vs Proposed (Short Term)

2030 Business-As-Usual Li Supply and Demand

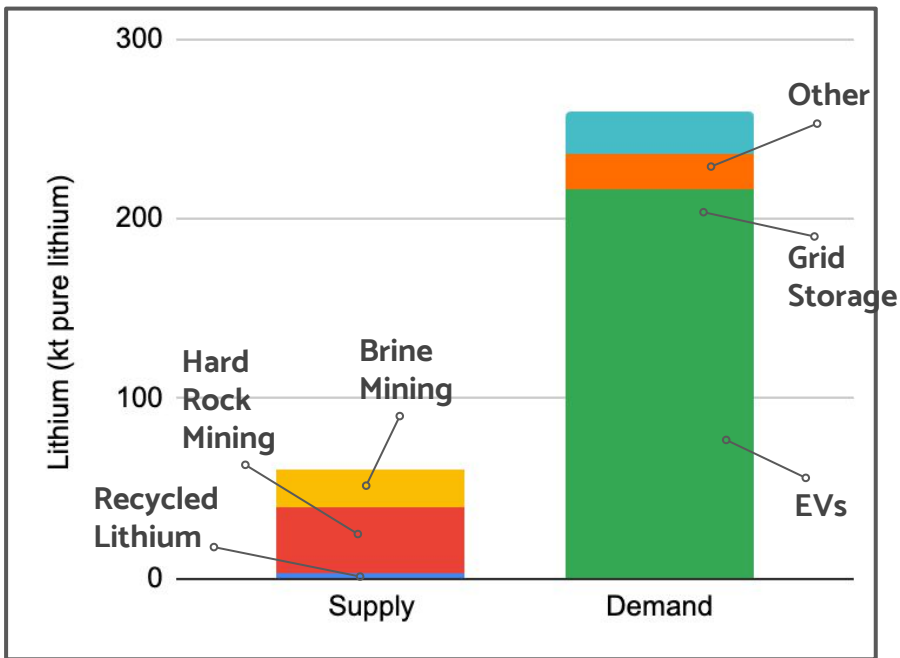


2030 Implemented Proposal Li Supply and Demand

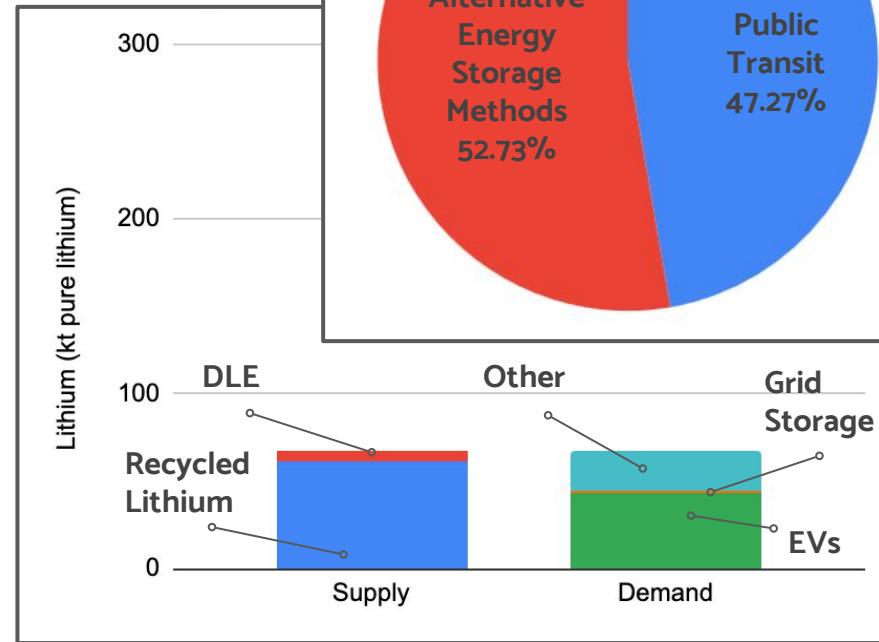


Current vs Proposed (Long Term)

2050 Business-As-Usual Li Supply and Demand



2050 Implemented and Demand



BIG THANKS

David McGee - *Terrascope Director,*

Ari Epstein - *Associate Director*

Michelle Contos - *Community Coordinator, &*

Maddie Paoletti - *Graduate Teaching Assistant*

Thank you to all of our UTFs, mentors, and experts!

Presentation template by [Slidesgo](#)

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